Sai Bhavani Gottumukkala,
Research Scholar,
Systems Biology,
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Education

Year	Degree/Exam	Institute	CGPA/Percentage
2018- Present	PhD	National Institute of Technology, Warangal	-
2018	Master of Technology	IIT Kharagpur	8.93/10
2016	Bachelor of Technology	National Institute of Technology, Warangal	8.14/10

Research Experience

2018, December - Present — Research Scholar
Thesis: Systems Modelling and Analysis Approach to Investigate the Cancer Signalling Pathways (**Supervisor**: Dr Anbumathi P, Department of Biotechnology, NIT Warangal, India)

2017-2018 — M. Tech Dissertation
Thesis: A Comparative Study of Protein-RNA Interface In Intrinsically Disordered Proteins and Ordered Proteins (**Supervisor**: Dr Ranjith Prasad Bahadur, Department of Biotechnology and Biochemical Engineering, IIT Kharagpur, India)

2015-2016 — B. Tech Dissertation
Thesis: Production of Bio-Diesel from jatropha oil (**Supervisor**: Dr K Narsimhulu, Department of Biotechnology, NIT Warangal, India)

May-July 2015 — Summer Trainee
Thesis: Identification of proteins from A*cinetobacter Radioresistens* MMC5 (**Supervisor**: Dr K V Jagannadham, Centre for Cellular and Molecular Biology, Hyderabad, India)

May-July 2014 — Summer Project
Title: Isolation Identification & Screening of Thermophilic Fungi (**Supervisor**: Dr K Narsimhulu, Department of Biotechnology, NIT Warangal, India)

Teaching Experience

Adhoc Faculty
June – December 2018
Department of Biotechnology, NIT Andhra Pradesh, India
Courses Taught: Immunology, Bionanotechnology, Bioinformatics

Publications

- **Sai Bhavani, G**., & Palanisamy, A. (2023). Network motifs and hypermotifs in TGFβ-induced epithelial to mesenchymal transition and metastasis. Frontiers in Systems Biology, 3. https://doi.org/10.3389/fsysb.2023.1099951
- **Bhavani, G.S**., Palanisamy, A. SNAIL driven by a Feed Forward Loop Motif Promotes TGFβ Induced Epithelial to Mesenchymal Transition. *Biomed Phys Eng Express*. 2022 Jun 14. doi: 10.1088/2057-1976/ac7896. Epub ahead of print. **PMID**: 35700712
- **Bhavani, G.S.,** & Anbumathi, P. (2021). Systems Modelling and Analysis of TGF-β induced Epithelial to Mesenchymal Transition in cancer Metastasis. *AJIR Abstracts, 8. DOI https://doi.org/10.21467/abstracts.109*

- **Sai Bhavani G**, Akilan M, Anbumathi Palanisamy. Systems Modelling and Analysis of Signalling Pathways in Hepatocellular Carcinoma (Manuscript Submitted to Elsevier, Gene: Under Review)
- **Sai Bhavani G**, Trivadi S Ganesan, Anbumathi Palanisamy. Comprehensive Molecular Interaction Map of TGFβ Induced Epithelial to Mesenchymal Transition in Breast Cancer. (Manuscript Submitted to NPJ Systems Biology & Applications: Under Review)
- **Sai Bhavani G**, Anbumathi Palanisamy. Epithelial to mesenchymal transition in Lung cancer is an (im)balance of signals induced by TGFβ signaling. (Manuscript under Preparation)
- **Sai Bhavani, G**., & Palanisamy, A. Cancer Metastasis: Multistability Exerted by p53 over TGFβ Induced Epithelial to Mesenchymal Transition (Manuscript under Preparation)

Book Chapters

- **G. Sai Bhavani,** Pushkar Malakar, Anbumathi Palanisamy. Role of synthetic biology to build a sustainable vaccine industry. (Submitted to "Bioreactor Design Concepts for viral Vaccine production, Elsevier Publications, Accepted)
- **G. Sai Bhavani** and Anbumathi P. (2022). Working Women's Health, Wellbeing: Issues and Concerns. *Cambridge Scholars Publishing.* ISBN (10): 1-5275-7770-8 ISBN (13): 978-1-5275-7770-1
- **G. Sai Bhavani**, Sumathi K, Lavanya S and Anbumathi Palanisamy. Interdisciplinary Mathematical modeling approaches for infectious diseases. Nova Science Publishers. ISBN: 978-1-68507-984-0

Conference Presentations/ Proceedings/ Poster

- **S. B. Gottumukkala** and A. Palanisamy, "Cancer Networks: Insights Into The Disease Mechanisms," *2023 IEEE 20th India Council International Conference (INDICON)*, Hyderabad, India, 2023, pp. 245-251, doi: 10.1109/INDICON59947.2023.10440959
- **Sai Bhavani G**, Akilan M, Anbumathi P. Comprehensive Network of Signaling Pathways in Hepatocellular Carcinoma and Network Analysis Reveals PTK2, GSK3β, and β-catenin to be Crucial for Cancer Progression. Computational Models of Life: From molecular biology to digital twins, workshop organized by European Molecular Biology Organization (EMBO), from 26th November 2023 to 1st December 2023, Spain. (Received Grants from EMBO and PerMEDCoE (Barcelona supercomputing center)).
- **Sai Bhavani G**, Anbumathi P. 2023. Hypermotifs in biological networks: TGFβ induced EMT as a case study. 4th international conference on "Frontiers in Computing and Systems (COMSYS 2023)" held during Oct 16-17th organized by IIT Mandi, Himachal Pradesh. (Manuscript accepted for publication in Springer LNNS series)
- **G Sai Bhavani**, Anbumathi P. 2022. Epithelial to mesenchymal transition in Lung cancer is an (im)balance of signals induced by TGFβ signaling. 12th European Conference on Mathematical and Theoretical Biology (ECMTB-2022)" held on 19.09.2022 to 23.09.2022 organised by European Society for Mathematical and Theoretical Biology (ESMTB) and the Society for Mathematical Biology (SMB), Heidelberg, Germany.
- **G Sai Bhavani,** Anbumathi P. 2022. Malignant Tumors in Women: A Panoramic View. INTERNATIONAL CONFERENCE ON 'WOMEN IN LEARNING AND LEADERSHIP' (IcWILL - 2022) held on 8th – 9th March 2022, organised by Internal Complaints Committee, NIT Warangal, India.
- **G Sai Bhavani,** Anbumathi P. 2021. Systems Modelling and Analysis to Explore the Role of p53 and Mdm2 in TGFβ Induced Epithelial to Mesenchymal Transition. IITM EBI Winter School 2021 , 6th – 9th December 2021, jointly organised by BioModels (EMBL-EBI), IBSE RBC-DSAI from Indian Institute of Technology Madras (IITM).

- **G Sai Bhavani,** Anbumathi P. 2021. Molecular Interaction Map of TGFβ Induced Epithelial to Mesenchymal Transition in Breast Cancer. 6th Disease Maps Community Meeting held on 29th – 30th November 2021 hosted by the Department of Systems Biology and Bioinformatics, University of Rostock, Germany.
- **G Sai Bhavani**, Anbumathi P. 2021. Exploring the Contributions of Women Scientists in Field of Systems Biology. International Conference on Women Empowerment in Science and Technology held on 8th – 10th March 2021, Organised by Women Cell, NIT Warangal, India
- **G Sai Bhavani**, Anbumathi P. 2021. Exploring the Contributions of Women Scientists in Field of Synthetic Biology. International Conference on Women Empowerment in Science and Technology held on 8th – 10th March 2021, Organised by Women Cell, NIT Warangal, India
- **G Sai Bhavani**, Anbumathi P. 2021. A Survey based analysis of Women's Health, Wellbeing: Issues and Concerns. International Conference on Women Empowerment in Science and Technology held on 8th – 10th March 2021, Organised by Women Cell, NIT Warangal, India
- **G Sai Bhavani**, Anbumathi P. 2021. Systems Modeling and Analysis of TGF - β Induced Epithelial to Mesenchymal Transition in Cancer Metastasis. National Conference on Biological, Biochemical, Biomedical, Bioenergy and environmental Biotechnology held on 29th -30th January, 2021, organized by Department of Biotechnology, NIT Warangal, India
- **G. Sai Bhavani**, Shervin Sugumaran, Anbumathi P. 2019. Quantification of Oxidative stress regulation of fission yeast S. pombe. IITM-EBI modelling workshop, IIT Madras in collaboration with EMBL, 2-5th December 2019, Chennai, Tamil Nādu, India
- **G Sai Bhavani**, Anbumathi P. 2019. Working Women's Health, Wellbeing: A Survey Based Study on Issues and Concerns. Role of Women in Technical Education (RWTE-2019) held on 8th – 9th March 2019, Organised as a part of Diamond Jubilee celebrations under TEQIP-III at NIT Warangal, India

Conferences/ Workshops /FDPs/ and Training Programs Attended

- Webinar on "**Scientific Writing Workshop"** held on 19th April 2023, organized by ACS Publications
- Online Workshop on "**Algorithms that transformed Molecular Biology**" held on 26th March 2023, organised by Indraprastha Institute of Information Technology Delhi (IIIT-Delhi), India
- Intensive course on "**Introduction to Synthetic Biology**" held on 2-15th March 2023, organised by IIT Madras in association with NPTEL, India
- Online symposium on "**Physics of Cells and Tissues**" held on 15th – 16th February 2023, organised by BSSE, IISc, Bangalore, India.
- Online International workshop on "**Genomes to Pathways (3rd Edition, 2022)**" held on 3rd-21st December, organised by Decode Life, India.
- Virtual workshop on "**Modeling Biological Systems"** held on 26th – 27th May 2022, organised by Center for Bioinformatics and Computational Biology, Ashoka University in Association with Department of Biotechnology, Ministry of Science & Technology, Government of India.
- Online FDP on "**Bioinformatics: Algorithms, Applications and Tools**" held during 16th – 25th May 2022, organised by E&ICT Academy, National Institute of Technology Warangal, in association with University College of Engineering, BIT Campus, Tiruchirapalli-620 024
- Webinar on "**Conducting a Literature Review**" held on 19th April 2022 organised by Clarivate Webinars, Endnote
- Webinar on "**Use of Online Digital Information Resources**" held on 16th September 2021 organised by Central Library, NIT Warangal, India

- Two-day workshop on "**Key Scientific Writing Strategies for building a strong Manuscript**" held on 10th,12th August 2021, organized by ACS Publications.
- "**Webinar on Artificial Intelligence and Machine Learning**" held on July 8th, 2021, Organised by MIDHANI in Commemoration of India@75.
- Annual workshop on "**Computational Cell Biology**" held on 21 -23 June 2021, Organised by R D Berlin center for cell analysis and Modelling, University of Connecticut School of Medicine.
- Webinar **on "Invincible Under Pressure"** held on 12th June 2021 organised by Department of Psychology, Stella Maris College, Chennai, Tamil Nādu, India.
- Five Day FDP on "**Effective Teaching and Learning Practices in Computational Biology**" held on 15th-19th March 2021, Organised by Department of Biotechnology in association with TLC, NIT Warangal, India.
- Three-day **"International Conference on Women Empowerment in Science and Technology" entitled "Exploring the Contributions of Women Scientists in Field of Systems Biology"** held on 8th – 10th March 2021, Organised by Women Cell, NIT Warangal, India.
- Two Day "**National Conference on Biology, Biochemical, Biomedical, Bioenergy and Environment Biotechnology**" held on held on 29-30 January 2021, Organised by Department of Biotechnology, NIT Warangal, India
- One week "**R Advanced Workshop**" held on 14th -21st November 2020 by VIT Vellore in association with FOSSEE, IIT Bombay, India.
- Webinar on "**Current Trends and Advancements in Computational Drug Design**" held on 6th November 2020 conducted by TAB, NIT Warangal, India
- Short term course on "**Applications of MATLAB in Mathematics Science and Engineering**" on 15-17 September 2020, St. Vincent clg of Engineering and Technology in association with The Institution of Engineers.
- One week **"Science leadership Workshop"** held on 22nd – 28th June 2020 by Central University of Punjab, Bathinda, India.
- "**IITM – EBI Modelling Workshop**" held on 2nd – 5th December 2019, Organised by IIT Madras in collaboration with EMBL, India
- "**Artificial Intelligence**" held on 19th - 23rd November 2019, Organised by AICTE Training and Learning, NIT Warangal, India.
- "**Technical Communication Skills: Drafting and Presenting Research Papers**" held on April 1st -5th 2019, Organised by Department of Humanities and Social Sciences under TEQIP-III, NIT Warangal, India.
- "Teaching and Learning of **Engineering Mathematics using PYTHON** through Hands-On Experience" held on 5th -10th March 2019, Organised by Department of Mathematics in association with TLC, NIT Warangal, India
- Five day workshop on "**Research Methodology & Scholarly Writing Skills (RMSWS -2019)**" held on 21st- 25th January 2019, Organised by SC-ST cell, Dr. B.R.Ambedkar Leraning centre, NIT Warangal, , India
- One Day Workshop on "**Protein Extraction and Purification from Animal Tissue**" at Bio Axis DNA research center Pvt.Ltd, Hyderabad, India.

MOOC Specializations and Certifications (Self-Initiated)

- Coursera: Introduction to Systems Biology, MATLAB for beginners, Python For beginners, Python Data Structures, Python to Access Web Data, Python Data Bases, Capstone: Retrieving, Processing, and Visualizing Data with Python,
- Network Analysis in Systems Biology – Coursera
- EndNote20 certification series - EndNote

Awards and Achievements

- **Received Grants from EMBO and PerMedCoE to attend the workshop**: 'Computational Models of Life: From molecular biology to digital twins' Workshop held from Nov 26th – Dec 1st, 2023 organized by the European Molecular Biology Organization (EMBO) in Spain
- **Certificate of Participation** in Essay Writing Competition on "Lifestyle for Environment" administered by Shastri Indo-Canadian Institute, New Delhi in 2023-24.
- **Winner** at the department level in the Research Confluence -24 Organised by NIT Warangal
- **Best Poster Presentation** in "National Conference on Biological, Biochemical, Biomedical, Bioenergy and environmental Biotechnology" held on 29th -30th January, 2021, organized by Department of Biotechnology, NIT Warangal, India
- **Winner** for the Essay Writing Competition in '**Quick heal CY-FI Arts Festival- 2019**' Organised by Vishwakarma University, Pune, India (https://www.slideshare.net/CYFIArts/es-10-authorgottumutkala-sai-bhavani)
- Qualified GATE exam in 2016 and 2018
- Received institute merit scholarship in B.Tech for securing higher rank in class (top 5%) for the academic year 2015-2016

Technical Knowledge

- Familiar with UNIX and Windows OS
- MATLAB, Python, XPPAUT and R Programming Languages
- Adept in employing various Systems Biology based tools including Cytoscape, Cell Designer, Cell Collective, GINsim and BooleanNet for network modelling and analysis.
- Comfortable in learning and utilising new tools and software for network modelling and analysis
- Molecular biology techniques, Cell Culture, Immunology, Genetic Engineering

Other Research and Administration activities

- Extended my assistance in the guidance of following M.Tech and B.Tech dissertations on works titled
    - Systems Modelling and Analysis of Signalling Pathways in Hepatocellular Carcinoma (M.Tech dissertation, 2022-2023)
    - Exploring Disease Networks of Colorectal Cancer (Summer Internship, May – July 2023)
    - Quantification of the Regulatory Signals among the Cell cycle and Epithelial Mesenchymal Transition in Cancer (M.Tech dissertation, 2021-2022)
    - Systems Modeling and Analysis of CREB pathway in Glioma (B.Tech dissertation, 2018-2019)
- Served as the representative of PhD scholars for the Department of Biotechnology, NIT Warangal during the period of May 2022 to August 2023.

Other Activities

My hobbies include Playing Veena, Singing, Painting, Quilling.

References

1. **Dr Anbumathi P**
   Assistant Professor
   Room No: 425, Department of Biotechnology
   National Institute of Technology Warangal
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   Telangana, India – 506004
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2. **Dr Asim Bikas Das**
   Associate Professor
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   National Institute of Technology Warangal
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